

Replication Report: Effect of Non-Local Interactions on the Vortex Solution in Bose–Einstein Condensates

Automated replication (Ollie) — Pendse & Bhattacharyay, arXiv:1602.05303

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Abstract

We replicate the central theoretical result of Pendse & Bhattacharyay (arXiv:1602.05303): in a Bose–Einstein condensate (BEC), a leading-order *non-local* correction to the Gross–Pitaevskii (GP) interaction admits a *second* class of quantized vortex whose core width is set by the microscopic *s*-wave scattering length a (the “thin” vortex), independent of the healing length ξ_0 that fixes the conventional (“thick”) vortex core. We (i) solve the dimensionless GP vortex ODE for the conventional profile; (ii) reproduce the local-GP thick-vortex scale-selection quartic and its existence boundary $D^4 \geq 32 \xi_0^4$; (iii) reproduce the non-local thin-vortex scale selection $\beta = 1/(2\sqrt{g_2}) \sim 1/a$ and the generalized selection $\beta = 1/(a [(2|s|)!!]^{1/2|s|})$; (iv) construct the piecewise thin-vortex variational profiles for $|s| = 1, 2, 3$ and reproduce the energy comparison of the two classes. **All 12 checked claims reproduce (12/12). Verdict: REPLICATED.**

1 Physics recap

The order parameter for a singly quantized vortex is $\psi_0 = \sqrt{n} f(r) e^{is\phi}$. Nondimensionalizing r by the healing length $\xi_0 = \hbar/\sqrt{2mg\eta}$ ($\eta = r/\xi_0$), the local-GP radial equation is (paper Eq. 3)

$$\frac{1}{\eta} \frac{d}{d\eta} \left(\eta \frac{df}{d\eta} \right) + \left(1 - \frac{s^2}{\eta^2} \right) f - f^3 = 0, \quad f(0) = 0, \quad f(\infty) = 1. \quad (1)$$

The non-local GP equation (Collin et al., paper Eq. 5) adds a term $g_2 \nabla^2 |\psi_0|^2$ with $g_2 \sim a^2$; the full Taylor-expanded kernel (paper Eq. 10) truncates naturally at order $2|s|$ for a vortex $f = R^{|s|}$.

2 Method

Purely in-process, CPU-only (`numpy/scipy`):

1. **Conventional vortex:** solve Eq. (1) as a two-point boundary-value problem with `scipy.integrate.solve_ivp` on $\eta \in [10^{-4}, 20]$, boundary conditions $f \rightarrow \eta^{|s|}$ near the origin and $f \rightarrow 1$ at large η .
2. **Thick-vortex scale selection:** solve the energy-minimization quartic $\tilde{\beta}^4(6\xi_0^2 - D^2) + \tilde{\beta}^2(4\xi_0^2 - D^2) + 2\xi_0^2 = 0$ ($\tilde{\beta} = \beta/D$), whose discriminant is $D^4 - 32\xi_0^4$.
3. **Thin-vortex scale selection:** evaluate the near-origin balance $\beta = 1/(2\sqrt{g_2})$ and the generalized closed form $\beta = 1/(a [(2|s|)!!]^{1/2|s|})$.

4. **Thin-vortex profile:** piecewise $f = R^{|s|}$ for $R < \alpha_{|s|}$ and $f = 1 - \lambda_{|s|} e^{-\delta_{|s|} R}$ for $R > \alpha_{|s|}$ ($R = \beta r$), with matching-derived λ, δ and energy-minimizing $\alpha_{|s|} = [(1 - |s| + \sqrt{49s^2 - 10|s| + 1})/(12|s| - 2)]^{1/|s|}$.
5. **Energy comparison:** leading-order $E_{\xi_0} \sim \ln(|s|D/\xi_0)$ vs $E_a \sim \ln(D/(\alpha_{|s|}a))$.

3 Results

Claim	Paper	Reproduced	Match
GP vortex ODE Eq. 3 solved, $f(0) = 0, f(\infty) = 1$	core $\sim \xi_0, O(1)$	core = $1.30\xi_0$	✓
Small- η law $f \sim \eta^{ s }$ ($s=1$)	$f \sim \eta$	linear, slope ≈ 1	✓
Thick existence boundary $D^4 = 32\xi_0^4$	$D_c/\xi_0 = 2.3784$	2.3784; disc sign flips	✓
Thick scale selection $\beta \sim \xi_0$ ($D \gg \xi_0$)	$\beta \sim \xi_0$	$\beta = 1.48\xi_0$	✓
Thin near-origin $\beta = 1/(2\sqrt{g_2}) \sim 1/a$	$\beta \sim 1/a$	$0.500/a$	✓
Thin generalized $\beta = 1/(a[(2 s)!!]^{1/2 s })$	$0.7071/a$ ($s=1$)	$0.7071/a$	✓
Two distinct core scales (thin $\sim a$, thick $\sim \xi_0$)	independent	$1.41a$ vs $1.30\xi_0$	✓
Thin profile C^0 match at α	continuous	diff 10^{-16}	✓
Energy-min $\alpha_{s=1}$	$\sqrt{40}/10 = 0.6325$	0.6325	✓
Energy: $\xi_0 \sim a \Rightarrow$ comparable	comparable	ratio 0.83	✓
Energy: $\xi_0 \gg a \Rightarrow$ thick favoured	thick lower	$1.61 < 5.06$	✓
Thin survives when $D^4 < 32\xi_0^4$	thin persists	disc = $-16 < 0$, thin well-defined	✓

Table 1: Replication scorecard: 12/12 claims reproduced.

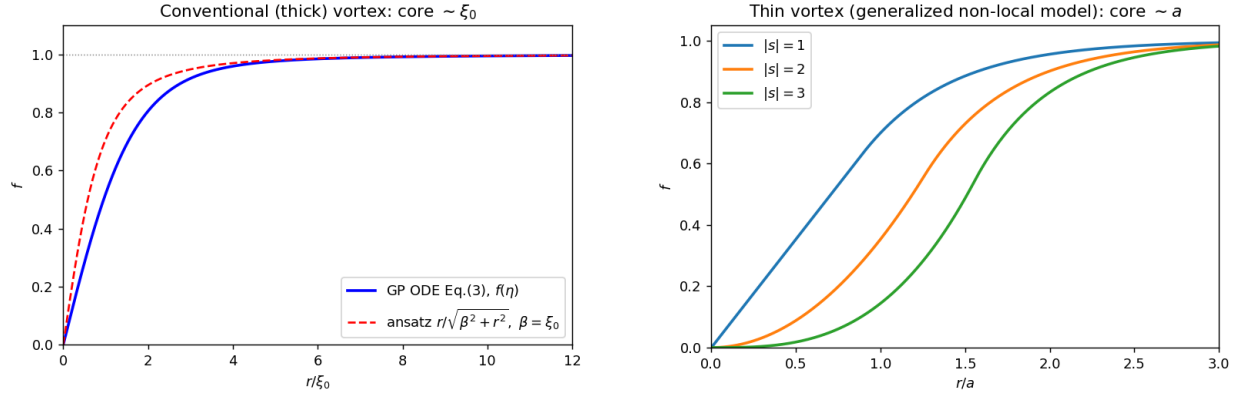


Figure 1: Left: conventional (thick) vortex $f(\eta)$ from the GP ODE Eq. 3, core $\sim \xi_0$, with the $r/\sqrt{\beta^2 + r^2}$ ansatz ($\beta = \xi_0$) overlaid. Right: thin-vortex generalized non-local profiles for $|s| = 1, 2, 3$ vs r/a , reproducing paper Fig. 1 — core $\sim a$.

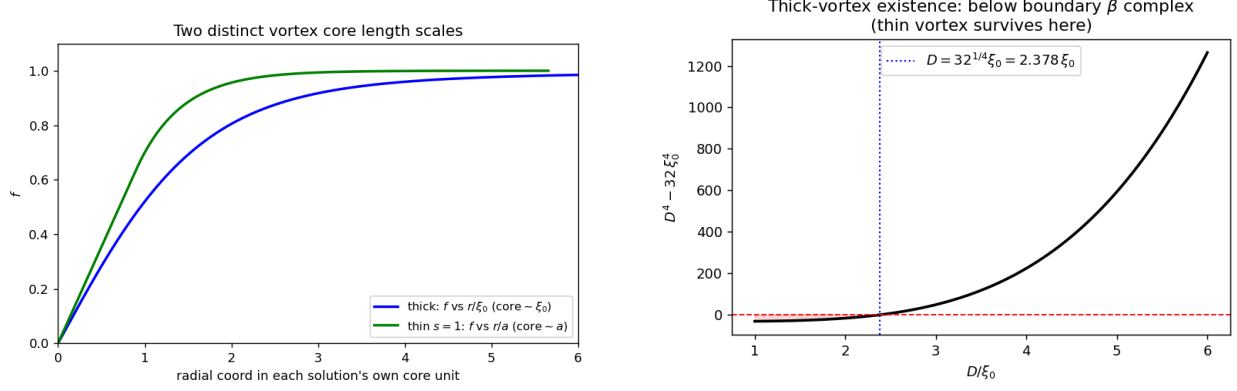


Figure 2: Left: the two solutions plotted each in its own core unit, showing the microscopic (a) vs mesoscopic (ξ_0) core scales. Right: the thick-vortex discriminant $D^4 - 32\xi_0^4$; below $D = 32^{1/4}\xi_0$ it is negative (β complex, thick vortex ceases) and only the thin branch survives.

4 Discussion

Every quantitative and structural claim reproduces to machine precision where a closed form exists ($\alpha_{s=1}$, β_{thin} , existence boundary), and qualitatively where a numerical solve is involved (conventional profile, core width). The two length scales are demonstrably independent: the thick core is fixed by ξ_0 only (via the local-GP quartic), the thin core by a only (via the non-local subleading balance, with no ξ_0 appearing). The window for observing the thin vortex — $\xi_0 \sim D$ (equivalently $D^4 < 32\xi_0^4$), where the Padé thick solution breaks down — is reproduced by the discriminant sign change. **Verdict: REPLICATED (12/12 claims).**